

### Assignment 1

1. *Time-dependent perturbation theory*: Consider interaction of a monochromatic light field (of the form  $E(t) = \hat{z}E_0(e^{i\omega t} + e^{-i\omega t})$ ) with an electron of an atom. Let us say that the light field interaction is present only between  $0 < t < t_o$  and that before  $t = 0$  the system is in some specific energy eigenstate  $|u_q^0\rangle$ . Using time-dependent perturbation theory, calculate the probability with which the system will make a transition to some other state.
2. *Time-dependent perturbation theory*: In class we calculated transition probability of an atom for a case when the amplitude of the perturbation induced by the applied light field was constant during the interval  $T$ . Redo this calculation for the following cases
  - (i) the amplitude is constant, but the light source is incoherent
  - (ii) the amplitude of the perturbation has a Gaussian profile with standard deviation  $T$ .
3. *Time-dependent perturbation theory*: For both constant (coherent and incoherent) and Gaussian cases discussed in problem 2, plot (3D plot) transition probability as a function of both detuning ( $\Delta\omega$ ) and time-duration of perturbation ( $T$ ) and draw inferences.
4. *Rabi oscillation*: In class we discussed interaction of a two-level atom with a classical light field. We considered a specific case of zero detuning for calculating occupation probabilities of states  $|1\rangle$  and  $|2\rangle$ . Derive and plot occupation probabilities of states  $|1\rangle$  and  $|2\rangle$  when detuning is non-zero. Assume the atom to be in state  $|1\rangle$  before the interaction began at  $t = 0$ .